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electronics
FOR YOU

Things You Wanted to Know!

Q1. What is the difference between a transformer and a coupled inductor?

A. Samiuddin
Through email

A1. A transformer is a static electrical device that transfers electrical energy between two or more circuits through electromagnetic induction. Depending on how a transformer is designed, you can use it to step down or step up voltages in power applications. It is also used for impedance matching or for isolating one part of a circuit from another.



Coupled inductor

Two inductors or coils that are linked by electromagnetic induction are said to be coupled inductors. These store energy and are used in tuned amplifiers and converters.

Q2. What is a nano PLC? Please give a PLC application.

Jayesh Wankhede,
Dhule, Maharashtra

A2. A programmable logic controller (PLC) is defined as an industrialised computer that includes the hardware and software required to automatically control production equipment, machinery, processes and other types of mechanical, electrical and electronic devices typically found in industries.

Nano PLCs are small, rugged PLCs that are the fastest growing areas of control logic. These provide economical solutions to basic control needs for simple machines. Nano PLCs have

less than 32 input/output functions and usually feature an all-in-one design that integrates an input module, output module and a logic module into a single unit that looks like a brick or a block.

A PLC represents a universal controller. It can be used for different applications and via the program installed in its memory. It provides the user with a simple means of changing, extending and optimising control processes.

Some applications of PLCs are:

Manufacturing processes. Lead-acid battery plants, extruder automation factories and silo-feeding control systems

Travel industry. Escalator operations, monitored safety control systems and lift operations

Aerospace. Water tank quenching systems

Printing industry. Offset Web press print register control systems and multi-stage screen washing systems

Textile industry. Industrial batch washing machine control systems and closed loop textile shrinkage systems

Q3. What is solar energy? Which component of the solar panel is used for conversion of solar energy to electrical energy?

Satyajit Samal
Through email

A3. Solar energy is the radiation and heat received from the Sun. The radiation is converted into electrical energy with the help of solar panels, which are made of solar cells.

A solar cell is a semiconductor device (p-n junction) that converts solar energy directly into electrical energy using photovoltaic effect. A cell is a small disk of a semiconductor like silicon. Multiple disks are attached to form a circuit using a wire. As sunlight strikes the semiconductor, light is converted into electricity that flows through the circuit. As soon as the light is removed, the solar cells stop producing power.

Answers compiled by EFY Lab engineers. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender